

LAST NAME: _____

Practice Quiz 04A — MATH 2055 — Fall 2026

Practice for Quiz 04

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. (a) Let A, B be sets. Prove $A \cup (B \setminus A) = A \cup B$ by a chain of equivalences beginning with membership in the left side. Name the definition or logical law used at each step.

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- (b) Let $\mathcal{F} = \{\{2\}, \{2, 5\}\}$. Compute $\bigcup \mathcal{F}$ and $\mathcal{P}(\bigcup \mathcal{F})$.